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To: The Annals of Applied Statistics

Cover letter

Dear Editors,

We are pleased to submit our article entitled «Increment Based Correlation» for your consideration.

This work introduces a new dependence measure, the Cheskidov Correlation (CIC or Cheskidov iCorrelation), which quantifies synchrony of directional increments in ordered data. The method addresses a fundamental gap in classical correlation theory by providing a consistent measure for nonlinear, oscillatory and complex time-series structures where standard Pearson's mean-based coefficients, Spearman's or Kendall's are known to perform inadequately. The main contributions and innovations of the paper are:

1. A new correlation coefficient based on normalized forward differences;
2. Theoretical characterization via the Cauchy–Schwarz framework;
3. Analytical comparison with classical correlation measures;
4. Demonstrations showing that CIC avoids spurious correlations typical for oscillatory or symmetric nonlinear datasets.

The results are original, have not appeared in any prior publication or conference proceedings, and the article is not under consideration elsewhere. All authors have read and approved the submission and confirm adherence to your journal Ethical Principles for Authors. The manuscript is not a revised or resubmitted version of a previously rejected submission.

If accepted, we will provide fully annotated code and data used to generate the article, examples and figures. All real or synthetic data used in the paper will be supplied as Supplementary Material in accordance with journal requirements. The Supplement is written in the same format as the article and contains complete technical details.

All co-authors are willing to serve as referees for future submissions.

We believe that the proposed methodology may be of interest to a broad statistical audience, particularly those working with nonlinear dependence, time-series analysis, signal analysis and measures of association. The paper could also be considered for the Discussion Paper track because it introduces a new dependence measure and addresses a longstanding conceptual issue in statistical methodology.

Thank you for your consideration.

Sincerely yours,

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